Physical Chemistry Cumulative Exam
March 3, 2006

The Clapeyron equation for phase changes is given by $(dp/dT) = \Delta S/\Delta V$.

1. Explain the meaning of the variables in this equation. Make sure to specify the quantities fully. E.g., simply saying that T is the temperature is insufficient; the temperature *of what*.

2. Calculate the melting temperature of ice under a pressure of 80 bar. Give the result in degrees Celsius. The density of ice near 0° C is 0.92 g cm^{-3} and that of liquid water is 1.00 g cm^{-3} (both practically independent of the pressure). The molar enthalpy of fusion (melting) of ice is 6.01 kJ/mol. (You should know the melting temperature of ice under 1 bar pressure). 1 bar=10^5 Pa.

Hints:
(A) **Keep the units throughout.** Results which are off by orders of magnitude because of mixing up cm^3 and m^3 or bar and Pa will be graded as F, irrespectively of the correctness of the logic.

(B) Make the calculation in steps
Step 1: Calculate the entropy change for a given quantity of water (1 mol or 1 g, it does not matter which)
Step 2: Calculate the volume change
Step 3: Use the fact that the melting temperature does not change dramatically with pressure, and thus even 80 bar pressure can be considered as a small change. Therefore, you can equate the derivative (dp/dT) with the *finite* change
[80 bar – 1 bar]/[T(80 bar) – T(1 bar)]